UNIT - V

Structural Walls, Non-Structural Elements & Ductility

Syllabus (as given):

Strategies in location of structural walls - sectional shapes - variations in elevation - cantilever walls without openings - failure mechanism of non-structures - effects of non-structural elements on structural system - analysis of non-structural elements - prevention of non-structural damage. Ductility: impact, requirements, assessment, factors affecting ductility - ductile detailing as per IS 13920:2016 - behaviour of beams, columns and joints in RC buildings during earthquakes.

PART A : STRUCTURAL WALLS (SHEAR WALLS)

Definition - Structural (shear) wall

A vertical RC wall designed to resist lateral forces in its own plane (shear, bending, axial) and to give the building stiffness, strength and drift control. Acts as a vertical cantilever fixed at the foundation.

- *Advantages: very high stiffness -> small drift -> less non-structural damage; large strength; efficient in tall buildings (10-40 storeys); can be ductile if detailed (IS 13920).*
- *Shear-wall buildings performed well in Chile 1960/2010 (but needs proper detailing: 2010 boundary-element failures), Bhuj 2001 and Mexico.*
- *Disadvantages: concentrate large forces on foundations (overturning), reduce architectural flexibility.*

1.1 Strategies in the location of structural walls

- *Place walls symmetrically about both axes in plan so that centre of rigidity ≈ centre of mass (minimise torsion).*
- *Locate walls along the periphery / as far from CR as possible to maximise torsional resistance; avoid a single core at one end.*
- *Distribute walls evenly in both orthogonal directions (at least 2 walls in each direction, ideally near the outer edges) - walls should not all be on one side.*
- *Walls must be continuous to the foundation (no discontinuity/transfer girders), with uniform thickness variation in elevation.*
- *Put walls in continuous lines with columns so that slab spans remain small; avoid walls along the weak axis only.*
- *Do not put lift/stair cores at one corner; if unavoidable, add balancing walls on the opposite side.*
- *Walls with openings: openings should be small, aligned vertically, away from the wall ends (boundary zones) - use coupling beams for large vertical rows of openings.*
- *Shear walls at ends of long plans are not advisable: they restrain thermal/shrinkage movement and attract cracking.*

- Provide adequate foundation for walls: large footing/raft/piles to resist overturning (tension at heel).

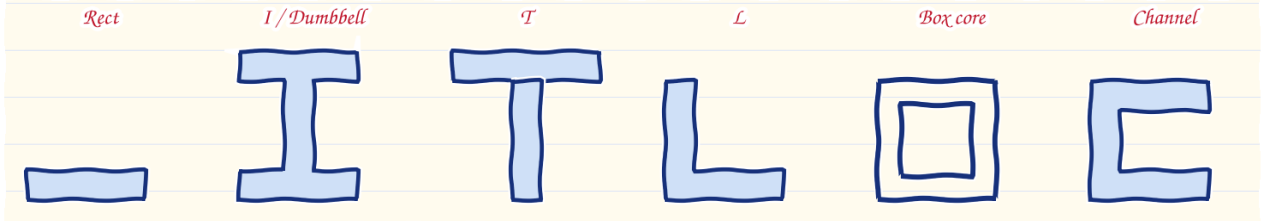


Fig: Sectional shapes of shear walls

1.2 Sectional shapes

Shape	Characteristics
Rectangular	Simplest; strong in own plane only, weak out-of-plane; used with columns as dual system
Barbell / I / dumbbell (flanged)	Boundary columns at wall ends act as flanges - increases flexural strength & stability (preferred where wall carries big moment)
T and L (flanged walls)	Flanges increase bending capacity/stiffness (effective flange width limited by half the distance to the next wall and a fraction of wall height, per code); torsionally unsymmetrical - careful
Channel / U-shaped	Used for lift/stair cores; open sections have low torsional stiffness, shear centre outside section
Box / closed core	Closed core has huge torsional and flexural stiffness; best for tall buildings; openings (doors) reduce stiffness; use coupling beams

1.3 Variations in elevation

- Wall thickness and reinforcement may be reduced gradually with height (shear and moment diminish upward) - avoid sudden changes (strength irregularity).
- Discontinuing a wall at a mid-level (floating wall) makes a weak/soft storey and large forces on transfer beams - prohibited by IS 1893:2016 for tall buildings without special design.
- Openings should occur in vertical rows - coupled shear walls (coupling beams) act as large frames: two walls + coupling beams -> energy dissipation via beam yielding (preferred over weak walls).
- Avoid large set-backs of wall in elevation; vary only gradually.
- Plastic hinge region at the base of wall (height l_w or $h_w/6$, whichever is more) is designed ductile; above it elastic.

1.4 Cantilever walls without openings

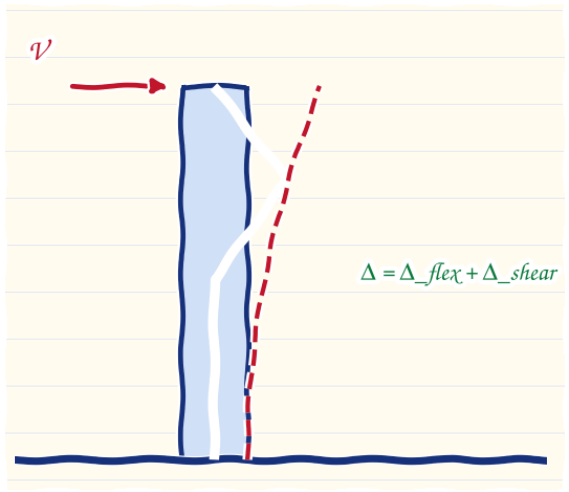


Fig: Cantilever wall: flexure + shear deflection

- Behaves as a vertical cantilever fixed at base: bending moment $M = \Sigma Q_i h_i$ (max at base), shear $V = \Sigma Q_i$ (max at base); axial load N from gravity.
- Slenderness (aspect ratio h_w/l_w): slender (> 2) -> flexure dominated, ductile if steel yields; squat (< 1) -> shear dominated, brittle (diagonal tension/sliding).
- Deflection at top: $\Delta = \Delta_{flex} + \Delta_{shear}$; for a tip load V: $\Delta = V h^3/(3EI) + 1.2\, V h/(GA)$ (for a triangular distribution of lateral load the flexural part is $\Delta_{flex} = 11\, V_B h^3/(60\, EI)$); $G = 0.4E$.
- Strength design: flexural design with P-M interaction (boundary elements), shear design with $V_u \le \tau_c, max$; ensure flexural capacity < shear capacity (capacity design): $V_u = V_{flex} = M_u/h_{eff}$ (× amplification).
- Modes of failure: flexural yielding (ductile), diagonal tension, diagonal compression (web crushing), sliding shear at construction joint, rocking/overturning, boundary-element crushing/buckling.

IS 13920:2016 - Ductile shear wall provisions (summary)

Thickness ≥ 150 mm; reinforcement ≥ 0.25% of gross area, both horizontally and vertically; if t > 200 mm or $V_u > 0.25\sqrt{f_{ck}}\, A_{cv}$ -> two curtains (layers) of steel.

Bar diameter ≤ t/10; spacing ≤ 450 mm (or 3t / l_w/5, least) both ways; vertical bars continue full height; horizontal bars anchored in boundary elements with 135° hook.

Boundary elements required when extreme-fibre compressive stress due to factored loads exceeds $0.2\, f_{ck}$; may be discontinued where stress < $0.15\, f_{ck}$. Boundary element: confining hoops as in columns (spacing ≤ 100 mm), length ≥ $0.15\, l_w$ (and ≥ wall thickness × 1.5), vertical steel 0.8% - 4%.

Shear: nominal shear stress $\tau_v = V_u/(t\, d_w) \le 0.63\sqrt{f_{ck}}$ (max); design shear is increased above analysis shear (capacity design / dynamic amplification for tall walls) so that flexure yields before shear fails.

Coupling beams ($l/d \le 4$): diagonal reinforcement cages in each direction, confined by ties.

Rule of thumb

Practical thumb rule: total shear-wall area in each direction ≈ 1.5% - 2.5% of floor area is enough in zones III-V; walls should be spread; dual systems (walls + SMRF) preferred.

PART B : NON-STRUCTURAL ELEMENTS (NSEs)

Definition - Non-structural elements

Components of a building that do not form part of the load-carrying system: partitions/infill, parapets, ceilings, facade/cladding, glazing, stairs (if isolated), mechanical-electrical-plumbing (MEP) equipment, water tanks, lifts, furniture, contents.

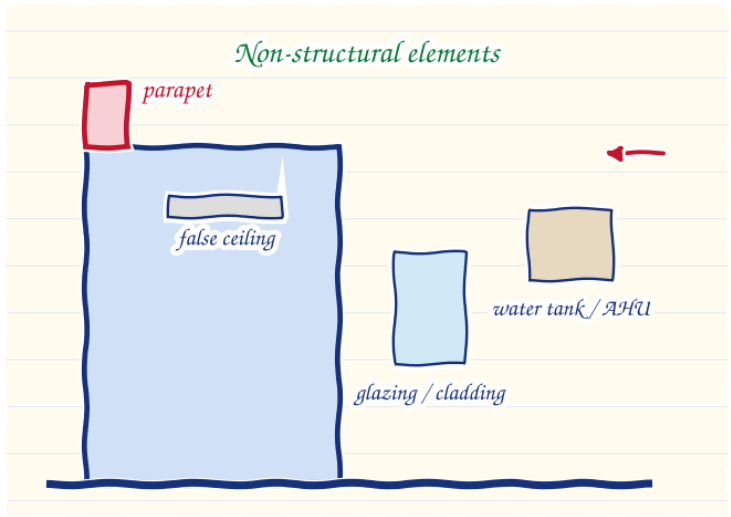


Fig: Typical non-structural elements

Significance: NSEs account for 70-85% of building cost (hospitals, offices) and cause most financial loss and injury (falling hazards), disruption (post-earthquake functionality). Even when the structure survives, NS damage can render a building unusable.

2.1 Failure mechanisms of non-structural elements

Element	Failure mechanism (cause)
Brick infill / partitions	In-plane: diagonal cracking due to storey drift; out-of-plane: overturning/bending - falling of walls
Parapets, chimneys, gables, balconies	Cantilever whiplash amplification at roof; overturning at base
Ceilings, light fittings	Loss of support, suspended grid failure; pounding with walls
Glazing / cladding	Drift imposed on rigid panels -> cracking/crushing; connection failure; falling glass
Heavy equipment/tanks/AHU	Sliding, overturning, loss of anchor; pipe rupture; leakage; fire
Stairs & lifts	Stairs act as diagonal braces -> stiffening -> failure; lift counterweight derailment; rails distortion
Contents / racks / furniture	Overturning, falling objects (books, servers)
Services (pipes, ducts, cables)	Fatigue, rupture at joints crossing seismic gaps

Two causes: (i) acceleration-sensitive (ceilings, equipment, parapets) - damage depends on floor acceleration; (ii) drift-sensitive (partitions, glazing, cladding, stairs) - depends on inter-storey drift.

2.2 Effects of non-structural elements on the structural system

- *Stiffening and strengthening: masonry infill raises stiffness 5-10x, reduces period (higher seismic force) and changes load path - may help (reduce drift) or hurt.*
- *Soft storey when infill is omitted in one floor (parking/entrance); short-column effect with partial infill; torsion from asymmetric infill (e.g., on 2 sides only).*
- *Added mass (heavy parapets, water tanks, partitions) increases base shear and top-heavy response; heavy roof tanks cause whiplash.*
- *Interference with structural detailing: rigid cladding/stairs hindering drift -> unintended forces on columns.*
- *IS 1893:2016: NS elements shall be accounted for in mass and stiffness (as struts) in analysis; separate them where they should not participate (gaps).*

2.3 Analysis of non-structural elements

Design lateral force on NS components/parts (IS 1893:2016 cl. 7.12 - concept)

$F_p = (Z/2)\,(I_p/R_p)\,a_p\,(1 + 2\,z/h)\,W_p$ (general form, as in most codes) - force increases with height z/h above base (up to 3 times at roof)

W_p = weight of the component; a_p = component amplification factor; R_p = component response modification factor; I_p = importance of component (1.0 / 1.5 for hazardous/essential)

Equivalent static force applied at the centre of mass of the component in the most unfavourable direction (horizontal), plus vertical.

- *Always use the factors a_p, R_p given in IS 1893:2016 Cl. 7.12 for the particular component - note them from the code tables in the exam; the idea is floor acceleration amplification x component weight.*
- *Vertical projections (parapets, balconies, canopies) are designed for about 5 times A_h on their weight (IS 1893).*
- *Drift check for drift-sensitive elements: relative displacement D_p = (drift ratio) x h_{storey}; glazing needs clearance $\delta \geq 1.25\,D_p$ (+ some); partition gaps ≥ computed drift.*
- *Equipment: designed to resist F_p, anchors (bolts, cast-in plates) checked for tension/shear with factor ≥ 1.0-1.5; flexible connections for piping.*

Worked example - parapet

Parapet 1.0 m high, 115 mm brick: weight per metre length W_p = 0.115 x 1.0 x 19 = 2.19 kN/m; building A_h = 0.10 (Zone IV); projection rule $F_p = 5\,A_h\,W_p$.

F_p = 5 x 0.10 x 2.19 = 1.09 kN/m acting at 0.5 m above base -> overturning moment = 1.09 x 0.5 = 0.55 kN.m/m

Resisting moment of self weight = 2.19 x 0.115/2 = 0.126 kN.m/m -> 0.126 < 0.55: an unreinforced parapet overturns.

Remedy: RC parapet / vertical bars anchored into slab, reduce height to ≤ 0.6 m, or brace.

2.4 Prevention of non-structural damage

- *Isolate drift-sensitive NSEs: gaps/sliding joints between infill and frame (isolated infill), flexible sealants, glazing with clearance, partition tracks.*
- *Anchor/brace acceleration-sensitive items: parapets (reinforced, ≤ 1 m), water tanks, AHUs, cable trays, suspended ceilings (diagonal bracing wires), shelving to walls.*
- *Reduce mass and keep it low; avoid heavy elements at the top.*
- *Positive connections: dowels/mesh to tie masonry infill to frame; bands in infill; anchors for cladding with slotted holes.*
- *Flexible service connections across seismic joints; shut-off valves; battery lighting; fire-stop.*
- *Limit drift to 0.004 h (IS 1893) to avoid NS damage; use stiffer structure for hospitals (I = 1.5).*
- *Quality control + regular inspection; code compliance for MEP (IS 1893 Part 1 cl 7.12; FEMA 74 guidelines).*

PART C : DUCTILITY CONSIDERATIONS IN EQ RESISTANT

3.1 Introduction and impact of ductility

- *A ductile structure deforms inelastically without strength loss; the area under the force-displacement loop = energy dissipated.*
- *Design philosophy: code forces are 1/R of elastic forces; the structure survives MCE only because of ductility (and over-strength).*
- *Impact of ductility: (i) reduces design force (R), (ii) prevents sudden collapse, (iii) redistributes moments, (iv) gives warning before failure (large cracks/deflection), (v) energy dissipation through hysteresis, (vi) allows structure to survive even when ground motion exceeds the design level.*

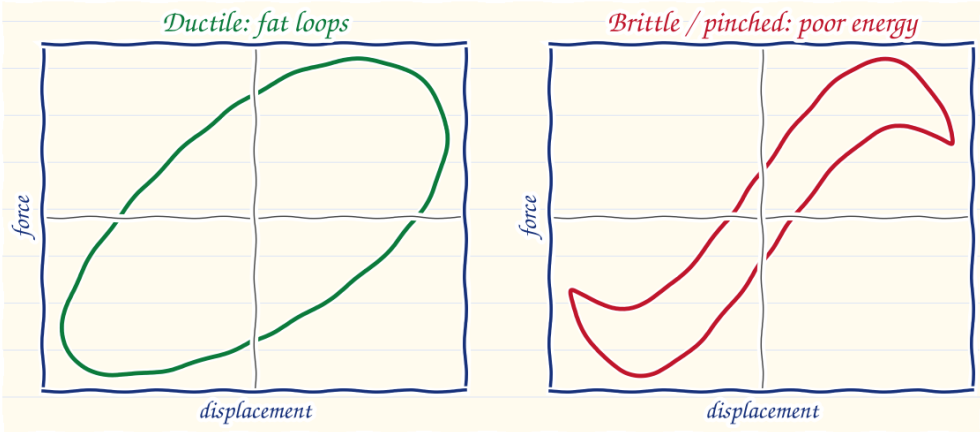


Fig: Hysteresis loops: ductile (fat) vs brittle (pinched)

3.2 Requirements for ductility

- *Material: Fe 415/500 (D) steel with elongation ≥ 14.5% and UTS/YS ≥ 1.15; concrete ≥ M20 (M25 in tall buildings, Zone III-V).*
- *Member level: under-reinforced beams (tension steel yields first), limited axial load ratio in columns ($P/(f_{ck} A_g) \leq 0.4$-0.5), confined concrete, adequate shear capacity.*
- *System level: weak beam-strong column ($\Sigma M_c \geq 1.4\, \Sigma M_b$), regular configuration, redundancy, no soft storey, ductile joints.*

- *Detailing: hooks 135°, closely spaced stirrups, adequate lap/anchorage, no splicing in plastic hinge zones.*
- *Plastic hinge rotation capacity: $\theta_p = (\varphi_u - \varphi_y) L_p$; $L_p \approx 0.5\,d + 0.05\,z$ (Paulay-Priestley) or $\approx D$.*

3.3 Assessment of ductility

Type	Definition	Typical
Displacement ductility	$\mu_\Delta = \Delta_u / \Delta_y$ (from pushover / load-deflection)	SMRF: 3-5
Curvature ductility	$\mu_\varphi = \varphi_u / \varphi_y$ (moment-curvature M-φ of section)	10-20 for good beam sections
Rotation ductility	$\mu_\theta = \theta_u / \theta_y$	related to hinge
Energy ductility	$E_{dissipated} / E_{elastic}$	Hysteresis area
Overall relation	$\mu_\Delta = 1 + 3(\mu_\varphi - 1)(L_p/L)(1 - 0.5\,L_p/L)$	Cantilever of length L

Moment-curvature quantities

$\varphi_y = \varepsilon_y / (d - x_u, y)$ ($\approx 2.1\,\varepsilon_y / D$ for beams); $\varphi_u = \varepsilon_{cu} / x_u = 0.0035 / x_u$ (IS 456 limit strain)
$\mu_\varphi = \varphi_u / \varphi_y$; tension steel increases x_u -> lowers μ_φ; compression steel and confinement raise μ_φ

Numerical

Example: beam with $\varphi_y = 5.5 \times 10^{-6}$ /mm and $\varphi_u = 5.0 \times 10^{-5}$ /mm -> $\mu_\varphi = 9.1$; with $L_p/L = 0.1$: $\mu_\Delta = 1 + 3(8.1)(0.1)(0.95) = 3.3$.

3.4 Factors affecting ductility

Factor	Effect on ductility
Tension steel ratio ρ	Increase in ρ reduces ductility (x_u rises; $\rho_{max} = 2.5\%$ in beams)
Compression steel ρ′	Increases ductility (reduces x_u); IS 13920: A_s,bottom ≥ 50% A_s,top at joint face
Axial load ratio	Higher axial load reduces ductility - limit $P/(f_{ck} A)$ in columns
Confinement (hoops, spacing, hook)	Closer ties & 135° hooks increase ductility strongly
Concrete strength	Higher f_{ck} -> more brittle (lower ε_{cu} in unconfined) - use M20-M30; avoid very high grades
Steel grade / ε_y	Lower yield strain (lower f_y) improves μ; strain-hardening ratio > 1.15 ideal; avoid CTD bars
Shear span / shear stress	Low shear span ratio (short members) -> brittle shear failure
Bond/anchorage and lap	Poor bond/lap in hinge region reduces ductility - use hooks & lap outside hinge zones
Cover and bar spacing	Spalling of cover removes confinement - small cover, tight cage

PART D : DUCTILE DETAILING AS PER IS 13920:2016

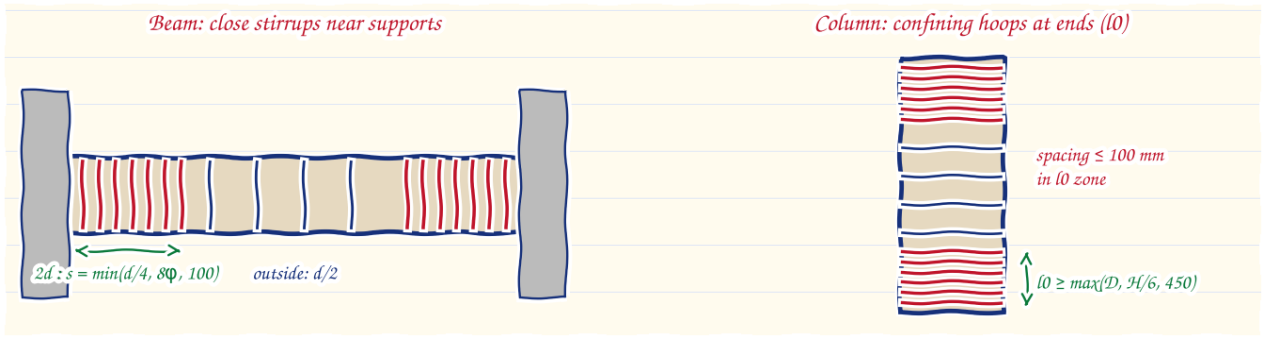


Fig: Confining reinforcement at member ends

4.1 Beams

- Geometry: width $b \geq 200$ mm; $b/D \geq 0.3$; depth $D \leq$ 1/4 clear span; .
- Longitudinal steel: min $\rho = 0.24\sqrt{f_{ck}}/f_y$; max $\rho = 2.5\%$; at least 2 bars at top and bottom through the full length; positive steel at joint face ≥ 50% of negative steel there; any section top/bottom ≥ 25% of max moment capacity at the joint face.
- Lap splices: only when hoops are provided over the lap length (spacing ≤ 150 mm); not within a joint, nor within 2d from joint face, nor at locations of yield; ≤ 50% bars lapped at one section.
- Web (stirrup) reinforcement: closed hoops with 135° hook, extension ≥ 10 d_b (≥ 75 mm); first hoop at 50 mm from the joint face; spacing in the hinge zone (2d from face) = min(d/4, 8 d_b, 100 mm); beyond that ≤ d/2; diameter ≥ 6 mm (8 mm if clear span > 5 m).
- Design shear: $V_e = V_{u,gravity} \pm 1.4\,(M_{u,lim}^{sag} + M_{u,lim}^{hog})/L$ (capacity design) - ensures flexural failure first.

4.2 Columns

- Geometry: min dimension ≥ 300 mm; ratio of shorter to longer side $b/D \geq 0.4$.
- Longitudinal steel: min 0.8% and max 4% of gross area (6% not allowed in lapped region); bars spaced ≤ 200 mm; lap splices in the central half of column height, with hoop spacing ≤ 150 mm over lap.
- Confining hoops at both ends over length $l_0 \geq \max(D, H_{clear}/6, 450\text{ mm})$ - spacing $s \leq$ 1/4 of min member dimension, and 75 mm ≤ s ≤ 100 mm; dia ≥ 8 mm; hoops with 135° hooks and extension ≥ 10 d_b; cross-ties/supplementary ties so that every corner bar and alternate bar is supported (no bar > 200 mm from a tied bar).
- Area of hoop: $A_{sh} = 0.18\,s\,h\,(f_{ck}/f_y)(A_g/A_k - 1)$, not less than $0.05\,s\,h\,(f_{ck}/f_y)$; circular spiral: $A_{sh} = 0.09\,s\,D_k\,(f_{ck}/f_y)(A_g/A_k - 1)$; h = longer dimension of rectangular hoop; A_k = core area.
- Remaining column height: spacing ≤ 6 d_b (smallest long. bar) and ≤ 150 mm.
- Strong column - weak beam: $\Sigma M_c \geq 1.4\,\Sigma M_b$ (2016) [1.1 in older code]. Design shear in columns: V_e from column plastic moments $V_e = 1.4\,(M_{u,col}^{top} + M_{u,col}^{bottom})/h_{st}$ (not less than analysis shear).
- Special confining continue through the joint and above/below beam up to l_0 (full height if column is short or below discontinuous walls).

4.3 Beam-column joints

- Joint = a vulnerable region with high shear ($V_{joint} = T_{beam} + C_{beam} - V_{col}$): joint shear failure causes collapse (Northridge, Kobe, Bhuj).
- Confine the joint with closed hoops at spacing ≤ 100 mm (same as column ends) - at least 2 layers of hoops through joint, with ≥ 135° hooks.
- Beam bars anchored in column with development length measured from the inner face of the column core; if column depth < required, bend bars into joint (hook 90° with 8 d_b length in the confined core).
- Interior joint: nominal shear stress limit; exterior joint: anchorage of top & bottom bars with 90° bends into core; corner joints with eccentric beams need larger columns.
- Where beam width > column width, extra hoops required; avoid eccentric beams > b/4.

PART E : BEHAVIOUR OF RC MEMBERS DURING EARTHQ

Member	Behaviour / typical damage	Prevention
Beams	Flexural hinges near column faces (ductile - desirable); cracks, spalling; shear failure (X-cracks) if stirrups sparse; bar slip/anchorage; torsion cracks in edge beams	Under-reinforced, closed stirrups at 2d, capacity shear, 135° hooks
Columns	Hinging at ends (spalling, bar buckling) = storey mechanism; shear failure (X-cracks) in short columns; crushing / axial failure under high axial load; lap-splice failure at base; bending-compression failure in corner columns	Strong column - weak beam, closely spaced hoops over l_0, limit axial load, no lap at ends, avoid short columns
Beam-column joints	Diagonal shear cracks across joint; loss of bond/anchorage (bar pull-out); joint failure -> collapse; exterior and corner joints worst	Joint hoops; proper anchorage; $\Sigma M_c \geq 1.4\,\Sigma M_b$
Slabs/diaphragms	Cracking at openings, re-entrant corners, weak connection to walls; punching shear failure in flat slabs (no beams)	Beam-column frame, trim bars, slab bands
Shear walls	Flexural hinge at base, boundary-element crushing, sliding shear, diagonal tension, coupling-beam failure	Boundary elements, 2 curtains, capacity design
Infill + frame	Infill cracking, short columns, soft storey	Isolated infill, full-height walls
Foundations	Differential settlement, liquefaction, pile failure	Ties, raft/piles, soil improvement

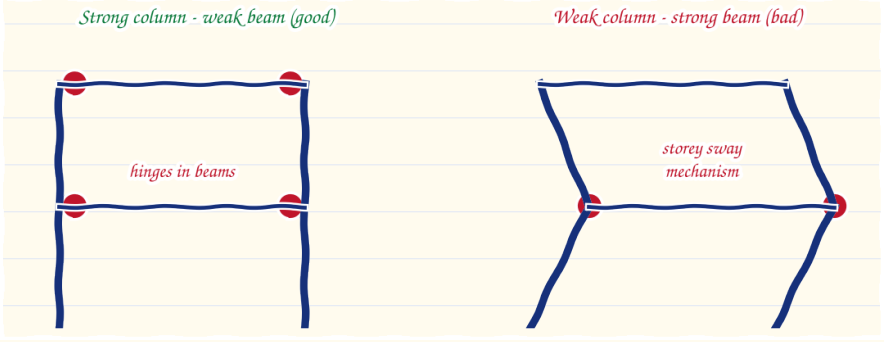


Fig: Desirable beam mechanism vs undesirable storey mechanism

- *Sequence of ideal failure: beam yielding in flexure -> beam hinges form at all floors -> large energy dissipation -> collapse only after large drifts. Avoid: shear failure of beams/columns, joint failure, anchorage failure, column hinging.*
- *Hysteretic behaviour: ductile members show stable fat loops; poorly detailed ones show pinching, strength decay and stiffness degradation after few cycles.*
- *Beams with deep sections and large span-to-depth give more ductility; short deep beams fail in shear.*
- *Columns: ductility drops sharply with high axial load; weak column failure leads to pancaking (Bhuj/Kobe).*

Quick example: Ash for column hoops

Column 400 x 400, cover 40 mm, hoop dia 10 mm -> core 320 x 320 (to hoop centre-line h = 320 mm); A_g = 160 000, $A_k = 320^2$ = 102 400 mm²

f_{ck} = 25, f_y = 415, S = 100 mm: $A_{sh} = 0.18 \times 100 \times 320 \times (25/415) \times (160000/102400 - 1) = 0.18 \times 100 \times 320 \times 0.0602 \times 0.5625$ = 195 mm²

Provide 10 mm hoops with 2 legs each way (2 x 78.5 = 157 mm²) -> add cross-tie / use 3 legs (235 mm²) ✓ or reduce spacing to 75 mm.

PART F : WORKED EXAMPLES

Ex.1 Is a boundary element needed? (shear wall)

Wall l_w = 4.0 m, t = 0.20 m, f_{ck} = 25 MPa; at base (factored): P_u = 3000 kN, M_u = 1200 kN.m

A = 0.8 m²; $Z = t\,l_w^2/6 = 0.2 \times 16/6 = 0.533$ m³

$\sigma_{max} = P_u/A + M_u/Z = 3750 + 2250 = 6000$ kN/m² = 6.0 MPa; $0.2\,f_{ck}$ = 5.0 MPa

6.0 > 5.0 -> boundary elements required at both ends; they can be stopped where $\sigma < 0.15\,f_{ck}$ = 3.75 MPa (up the height).

Ex.2 Beam ductile detailing numbers (IS 13920:2016)

Beam 300 x 500 mm, d = 450 mm, smallest long. bar 16 mm, f_{ck} = 25, f_y = 415

$\rho_{min} = 0.24\sqrt{25/415} = 0.289\%$ -> $A_{st,min} = 0.00289 \times 300 \times 450$ = 390 mm²; ρ_{max} = 2.5% -> 3375 mm²; b/D = 0.6 ≥ 0.3 ✓

Hoop zone length = 2d = 900 mm at each end; spacing = min(d/4 = 112, 8 x 16 = 128, 100) = 100 mm; first hoop at 50 mm -> hoops at 50, 150, ... 850 mm (9 hoops)

Outside the zone: spacing ≤ d/2 = 225 -> provide 200 mm. Hook: 135° with extension ≥ 10 x 8 = 80 mm (≥ 75 mm) for 8 mm hoops.

Ex.3 Joint shear (capacity-based, approx.)

Beam top steel 3 - 20 mm (942 mm²), bottom 3 - 16 mm (603 mm²); steel stress 1.25 f_y = 519 MPa (overstrength)

T_1 = 519 × 942 = 489 kN; C_2 = 519 × 603 = 313 kN; column shear V_{col} ≈ 130 kN

$V_{joint} = T_1 + C_2 - V_{col}$ = 489 + 313 - 130 = 672 kN; joint area 400 × 400 -> τ = 672e3/160000 = 4.2 MPa (compare with code limit ≈ 1.2-1.5 $\sqrt{f_{ck}}$ = 6-7.5 MPa) ✓ - provide joint hoops @ 100 mm.

Ex.4 R from ductility

If displacement ductility available μ = 4: equal displacement rule (T > 0.5 s): R_μ = 4; equal energy (short period): $R_\mu = \sqrt{(2 \times 4 - 1)}$ = 2.65

With over-strength Ω = 1.6 -> $R = R_\mu \Omega$ = 4 × 1.6 = 6.4 (long period); codes adopt conservative R = 5 for SMRF.

COMPARISON & SUMMARY TABLES

Item	Ordinary (non-ductile) detailing	IS 13920:2016 ductile detailing
Stirrups	90° hooks, 150-300 mm spacing	135° hooks, 100 mm in hinge zones, first at 50 mm
Column ends	Normal ties	Special confining hoops over l_0, s ≤ 100 mm
Laps	Anywhere	Only in central half of columns / away from joints, hoops over lap
Joint	No ties or few	Hoops continued through joint
Beam steel	ρ limit 4%	ρ_{max} 2.5%, 2 bars through, top-bottom ratios
$\Sigma M_c / \Sigma M_b$	Not required	≥ 1.4 (1.1 in older version)
Shear design	Analysis shear	Capacity (plastic hinge) shear
Steel	Any Fe 415	Fe 415/500/550 (D) with ductility limits

Element	Key IS 13920:2016 numbers
Beam	b ≥ 200; b/D ≥ 0.3; D ≤ L/4; ρ: 0.24$\sqrt{fck/fy}$ to 2.5%; hoop zone 2d; s = min(d/4, 8φ, 100); first at 50
Column	min 300; b/D ≥ 0.4; ρ 0.8-4%; l_0 = max(D, H/6, 450); s ≤ b/4, 75-100; laps central half
Joint	hoops s ≤ 100 through joint; anchorage from inner core face
Wall	t ≥ 150; ρ ≥ 0.25% each way; 2 curtains if t > 200; boundary element if σ > 0.2 fck
Concrete / steel	M20 min (M25 for H > 15 m in zones III-V); Fe 415/500/550(D), elongation ≥ 14.5%

Capacity design - the idea in 5 steps

1. Choose the desired yielding mechanism (beam hinges + base-of-column hinges).
2. Design the yielding members (beams) for flexure; provide ductile detailing.
3. Compute their probable/overstrength capacities (1.25 f_y).

4. Design all other members (columns, joints, shear in beams, foundations) for forces corresponding to those capacities.
5. Detail: hoops, anchorages, laps outside hinge zones.

QUICK REVISION - Likely questions

1. Strategies for the location of shear walls; sectional shapes; effect on torsion.
2. Cantilever shear wall behaviour; slender vs squat; IS 13920 provisions for walls and boundary elements.
3. Non-structural elements: failure mechanisms, effect on structural behaviour, analysis and prevention of damage.
4. Define ductility; types and relationships; factors affecting ductility; why ductility is needed.
5. Ductile detailing of beams and columns as per IS 13920:2016 (with sketches); special confining reinforcement.
6. Behaviour of beams, columns and joints during earthquakes; beam vs storey mechanism.